

An absolutely continuous trace formula for strongly commuting normal operators

Run `fa_banach_001`, lane 9
`agent_lane_09`

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Status. Candidate partial result, likely valid. It answers both the measure-existence and absolute-continuity parts of the source problem under the additional hypothesis that the two normal operators commute with one another.

1 Source question

Aleksandrov, Peller, and Potapov [1] ask the following in Open Problem 2 on page 6. If normal operators N_1, N_2 have trace-class difference, do finite Borel measures ν_1, ν_2 on $\mathbb{R}^2$ exist such that

$$\mathrm{Tr}(f(N_2) - f(N_1)) = \int_{\mathbb{R}^2} f_x d\nu_1 + \int_{\mathbb{R}^2} f_y d\nu_2 \quad (1)$$

for sufficiently smooth f ? If so, may the measures be chosen absolutely continuous with respect to planar Lebesgue measure?

norm $\|f(A) - f(B)\|_{\mathcal{S}_1}$. A natural question arises of whether the same can be said in the case of functions of two noncommuting self-adjoint operators. It would be interesting to obtain a kind of an optimal estimate for $|\text{trace}(f(A_2, B) - f(A_1, B))|$. This could lead to finding a version of a trace formula.

2. A trace formula for functions of normal operators. In §3 of this paper we have shown that the Lifshits–Krein trace formula (1.1) cannot be generalized to the case of pairs of noncommuting self-adjoint operators. What is the situation in the case of functions of pairs of commuting self-adjoint operators?

Clearly, it is the same as to consider functions of normal operators. Recall that in the paper [APPS] it was shown that the inequality

$$\|(f(N_2) - f(N_1))\|_{\mathcal{S}_p} \leq \text{const} \|f\|_{B_{\infty,1}^1} \|N_2 - N_1\|_{\mathcal{S}_p}$$

holds for any function f of Besov class $B_{\infty,1}^1(\mathbb{R}^2)$ and for any p in $[1, \infty]$; moreover, the constant does not depend on p .

The question is the following: let N_1 and N_2 be normal operators on Hilbert space. Is it true that there exist finite Borel measures ν_1 and ν_2 such that the trace formula

$$\text{trace}(f(N_2) - f(N_1)) = \int_{\mathbb{R}^2} \frac{\partial f}{\partial x}(x, y) d\nu_1(x, y) + \int_{\mathbb{R}^2} \frac{\partial f}{\partial y}(x, y) d\nu_2(x, y)$$

holds for sufficiently smooth functions f on $\mathbb{R}^2$? Note that if the answer to this question is positive, then for such normal operators N_1 and N_2 the inequality

$$|\text{trace}(f(N_2) - f(N_1))| \leq \text{const} \|f\|_{\text{Lip}}$$

holds for sufficiently nice functions f on $\mathbb{R}^2$. In any case, it would be important to find out whether under the above assumptions one can estimate $|\text{trace}(f(N_2) - f(N_1))|$ better than $\|(f(N_2) - f(N_1))\|_{\mathcal{S}_1}$.

If the above question has an affirmative answer, it is natural to ask the question of whether one can select measures ν_1 and ν_2 to be absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^2$.

Figure 1: Open Problem 2 in arXiv:1901.09495, page 6.

The theorem below gives an affirmative answer to both questions when $N_1 N_2 = N_2 N_1$. This is stronger than merely asking that the real and imaginary parts of each individual N_j commute; that latter condition is automatic from normality and is part of the unrestricted problem.

2 Proof intuition

Cross-commutation makes the trace-class perturbation $K = N_2 - N_1$ a compact normal operator commuting with N_1 . Every nonzero eigenspace of K is finite-dimensional and reduces N_1 . On the κ -eigenspace, diagonalising N_1 turns N_2 into the same diagonal matrix shifted by κ . The entire trace is therefore a summable collection of scalar endpoint differences $f(b) - f(a)$, whose total travel length is exactly $\|K\|_{\mathcal{S}_1}$.

The obvious representation integrates ∇f along the segment from a to b , but its measure is singular. To get absolute continuity, replace the segment by a compactly supported planar flow. First spread each endpoint to a smooth blob at scale $|b - a|$, using a radial field with one point source. Then transport one blob to the other along the segment. The resulting L^1 vector field $V_{a,b}$ has $\operatorname{div} V_{a,b} = \delta_a - \delta_b$ and $\|V_{a,b}\|_1 \leq 3|b - a|$. Summability in trace norm now permits all these absolutely continuous flows to be added.

3 A planar endpoint field

We identify $\mathbb{C}$ with $\mathbb{R}^2$, write m_2 for planar Lebesgue measure, and use the Euclidean norm for vector fields.

Lemma 1. *For every $a, b \in \mathbb{R}^2$, there is a compactly supported real vector field $V_{a,b} \in L^1(\mathbb{R}^2; \mathbb{R}^2)$ such that*

$$\operatorname{div} V_{a,b} = \delta_a - \delta_b \quad \text{in distributions}, \quad \|V_{a,b}\|_{L^1} \leq 3|b - a|.$$

Consequently, for every $f \in C^1(\mathbb{R}^2)$,

$$f(b) - f(a) = \int_{\mathbb{R}^2} \nabla f(x) \cdot V_{a,b}(x) dm_2(x). \quad (2)$$

Proof. The case $a = b$ is immediate. Fix a nonnegative radial $\eta \in C_c^\infty(B(0, 1))$ with $\int \eta dm_2 = 1$, and set

$$M(r) = \int_{|x| \leq r} \eta(x) dm_2(x), \quad G(x) = \frac{1 - M(|x|)}{2\pi} \frac{x}{|x|^2} \quad (x \neq 0).$$

Since $M(r) = 1$ for $r \geq 1$, the field G is compactly supported. Radial integration gives

$$\|G\|_1 = \int_0^1 (1 - M(r)) dr \leq 1.$$

The flux of G through a small circle about zero tends to one, while away from zero the radial divergence formula gives $\operatorname{div} G = -\eta$. Hence

$$\operatorname{div} G = \delta_0 - \eta \quad (3)$$

distributionally.

Put $h = b - a$, $r = |h|$, $\eta_r(x) = r^{-2}\eta(x/r)$, and $G_r(x) = r^{-1}G(x/r)$. Define

$$\begin{aligned} D(x) &= G_r(x - a) - G_r(x - b), \\ H(x) &= h \int_0^1 \eta_r(x - a - th) dt, \\ V_{a,b}(x) &= D(x) + H(x). \end{aligned}$$

Scaling (3) yields

$$\operatorname{div} D = \delta_a - \eta_r(\cdot - a) - \delta_b + \eta_r(\cdot - b).$$

Moreover,

$$\operatorname{div} H = \int_0^1 h \cdot \nabla \eta_r(x - a - th) dt = \eta_r(x - a) - \eta_r(x - b).$$

Thus $\operatorname{div} V_{a,b} = \delta_a - \delta_b$. Both fields are compactly supported, and

$$\|D\|_1 \leq 2r\|G\|_1 \leq 2r, \quad \|H\|_1 \leq |h| \int_0^1 \|\eta_r\|_1 dt = r.$$

This proves the norm estimate. Finally, distributional integration by parts gives $-\int V_{a,b} \cdot \nabla f = f(a) - f(b)$, which is (2). A cutoff equal to one on the compact support of the field justifies the same identity when f is not compactly supported. $\square$

4 The partial trace formula

Theorem 1. *Let N_1, N_2 be bounded normal operators on a complex Hilbert space. Assume*

$$N_1 N_2 = N_2 N_1, \quad K := N_2 - N_1 \in \mathcal{S}_1.$$

Then there is a real vector field $v = (v_1, v_2) \in L^1(\mathbb{R}^2; \mathbb{R}^2)$ such that

$$\operatorname{Tr}(f(N_2) - f(N_1)) = \int_{\mathbb{R}^2} (f_x v_1 + f_y v_2) dm_2 \quad (4)$$

for every complex-valued $f \in C^1(\mathbb{R}^2)$. The operator difference in (4) is trace class, and the densities satisfy

$$\int_{\mathbb{R}^2} |v| dm_2 \leq 3\|K\|_{\mathcal{S}_1}, \quad \|v_1\|_1 + \|v_2\|_1 \leq 3\sqrt{2}\|K\|_{\mathcal{S}_1}. \quad (5)$$

In particular, $\nu_j = v_j m_2$ are finite real signed Borel measures, absolutely continuous with respect to m_2 , and they solve the source trace formula in this subcase.

Proof. Because commuting bounded normal operators doubly commute by Fuglede's theorem [4], K is normal and commutes with both N_1 and N_1^* . Let $\{\kappa\}$ be the nonzero eigenvalues of the compact normal operator K , and let $E_\kappa = \ker(K - \kappa I)$. These spaces are finite-dimensional, mutually orthogonal, and reduce N_1 . Their orthogonal complement is $\ker K$, where $N_2 = N_1$.

On each E_κ , choose an orthonormal basis diagonalising the normal restriction of N_1 ; write its eigenvalues as $\lambda_{\kappa,1}, \dots, \lambda_{\kappa,m_\kappa}$. In the same basis, the eigenvalues of N_2 are $\lambda_{\kappa,j} + \kappa$. Trace-class summability says

$$\sum_{\kappa \neq 0} m_\kappa |\kappa| = \|K\|_{\mathcal{S}_1}. \quad (6)$$

For every pair $a = \lambda_{\kappa,j}$, $b = \lambda_{\kappa,j} + \kappa$, take the field from Lemma 1, and set

$$v = \sum_{\kappa \neq 0} \sum_{j=1}^{m_\kappa} V_{\lambda_{\kappa,j}, \lambda_{\kappa,j} + \kappa}.$$

By the lemma and (6), this series converges absolutely in $L^1(\mathbb{R}^2; \mathbb{R}^2)$, and its norm is at most $3\|K\|_{\mathcal{S}_1}$. The second bound in (5) follows from $|v_1| + |v_2| \leq \sqrt{2}|v|$.

All endpoints and field supports lie in a fixed compact set depending only on $\|N_1\|$ and $\|K\|$. Hence ∇f is bounded there. The scalar mean-value estimate and (6) show that the block diagonal operator $f(N_2) - f(N_1)$ is trace class. Absolute convergence, followed by Lemma 1, now gives

$$\begin{aligned}\mathrm{Tr}(f(N_2) - f(N_1)) &= \sum_{\kappa \neq 0} \sum_{j=1}^{m_\kappa} [f(\lambda_{\kappa,j} + \kappa) - f(\lambda_{\kappa,j})] \\ &= \int_{\mathbb{R}^2} \nabla f \cdot v \, dm_2,\end{aligned}$$

which is (4). □

5 Verification, limitations, and novelty status

The proof is deterministic and has no computational dependency. The two points most worth checking are the sign and scaling in $\mathrm{div} V_{a,b} = \delta_a - \delta_b$, and the use of Fuglede’s theorem to make each nonzero eigenspace of K reducing for N_1 . The construction also works on nonseparable Hilbert spaces: the trace-class range is separable and the complement lies in $\ker K$.

The cross-commutation hypothesis is a serious limitation. For arbitrary normal matrices A, B , the best possible trace bound over real one-Lipschitz f is the unnormalised planar W_1 matching cost between their eigenvalue multisets. Thus a dimension-free estimate of that cost by $\|A - B\|_{\mathcal{S}_1}$ would provide the missing finite-dimensional input; an unbounded sequence of ratios could instead feed a direct-sum counterexample. Hoffman–Wielandt controls W_2 by the Hilbert–Schmidt norm, but its conversion to W_1 loses a factor $\sqrt{n}$. The present proof does not resolve this hinge.

A bounded literature search through 11 August 2026 covered the exact problem wording, normal-operator trace formulas, commuting self-adjoint tuples, spectral-shift measures in two variables, and normal-matrix matching in Schatten norms. Chattopadhyay–Sinha [3] prove a two-variable Stokes-like formula for a second-order expression under Hilbert–Schmidt perturbations; it is not the first-order formula (1). Aleksandrov–Peller–Potapov–Sukochev [2] prove Besov/Schatten estimates for normal perturbations, but not the requested Lipschitz trace estimate. No exact statement of Theorem 1 was found. Novelty confidence is moderate rather than high because the argument becomes elementary after imposing cross-commutation.

Human-review recommendation. Retain this as a substantial partial theorem if the explicit L^1 endpoint field and the block trace argument verify. Do not treat it as resolving the arbitrary-normal question. The most valuable next step is the $\mathcal{S}_1$ -to- W_1 spectral matching hinge identified above.

References

- [1] A. B. Aleksandrov, V. V. Peller, and D. S. Potapov, *On a trace formula for functions of noncommuting operators*, arXiv:1901.09495 (2019).
- [2] A. B. Aleksandrov, V. V. Peller, D. Potapov, and F. Sukochev, *Functions of normal operators under perturbations*, Adv. Math. 226 (2011), 5216–5251; arXiv:1008.1638.
- [3] A. Chattopadhyay and K. B. Sinha, *Trace formula for two variables*, arXiv:1402.0792 (2014).
- [4] B. Fuglede, *A commutativity theorem for normal operators*, Proc. Natl. Acad. Sci. USA 36 (1950), 35–40.